

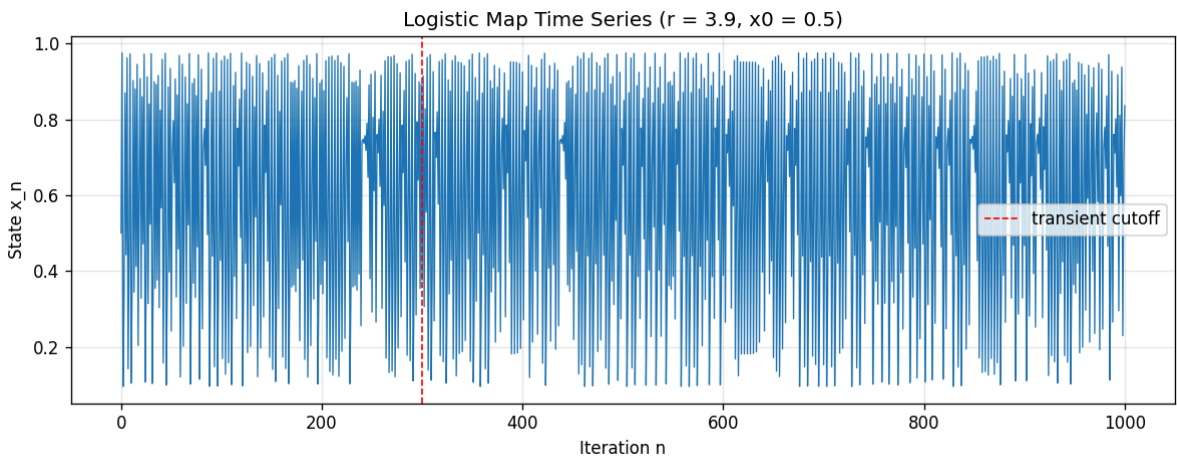
Task 1: Simulation and Sensitivity Analysis of Chaotic Systems

Part A: Discrete Chaotic Systems

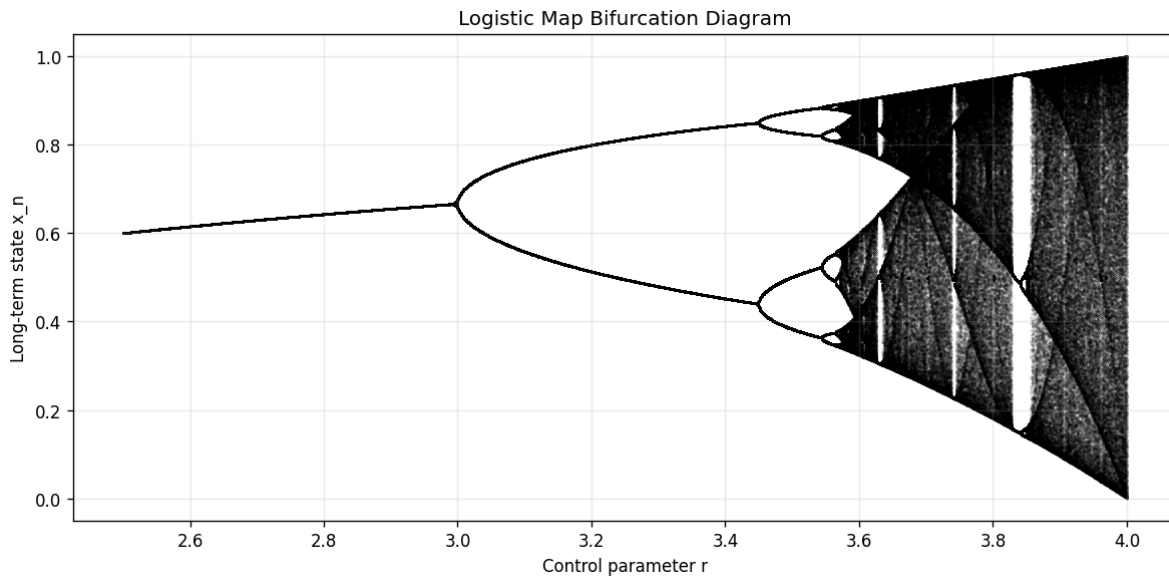
A.1 Logistic Map

$$x[n+1] = r * x[n] * (1 - x[n])$$

Setting	Value
State variable	x, with x in [0, 1]
Control parameter	r
Time-series r	3.9
Initial condition	x0 = 0.5
Iterations	1000
Transient ignored	first 300 iterations
Bifurcation range	r from 2.5 to 4.0
Bifurcation step	0.001
Bifurcation iterations	1200 total, first 500 ignored



From the plot, we can see that the logistic map does not settle into one repeated value when $r = 3.9$. After ignoring the first 300 iterations as transient, the sequence keeps jumping in an irregular way, but it stays between 0 and 1.



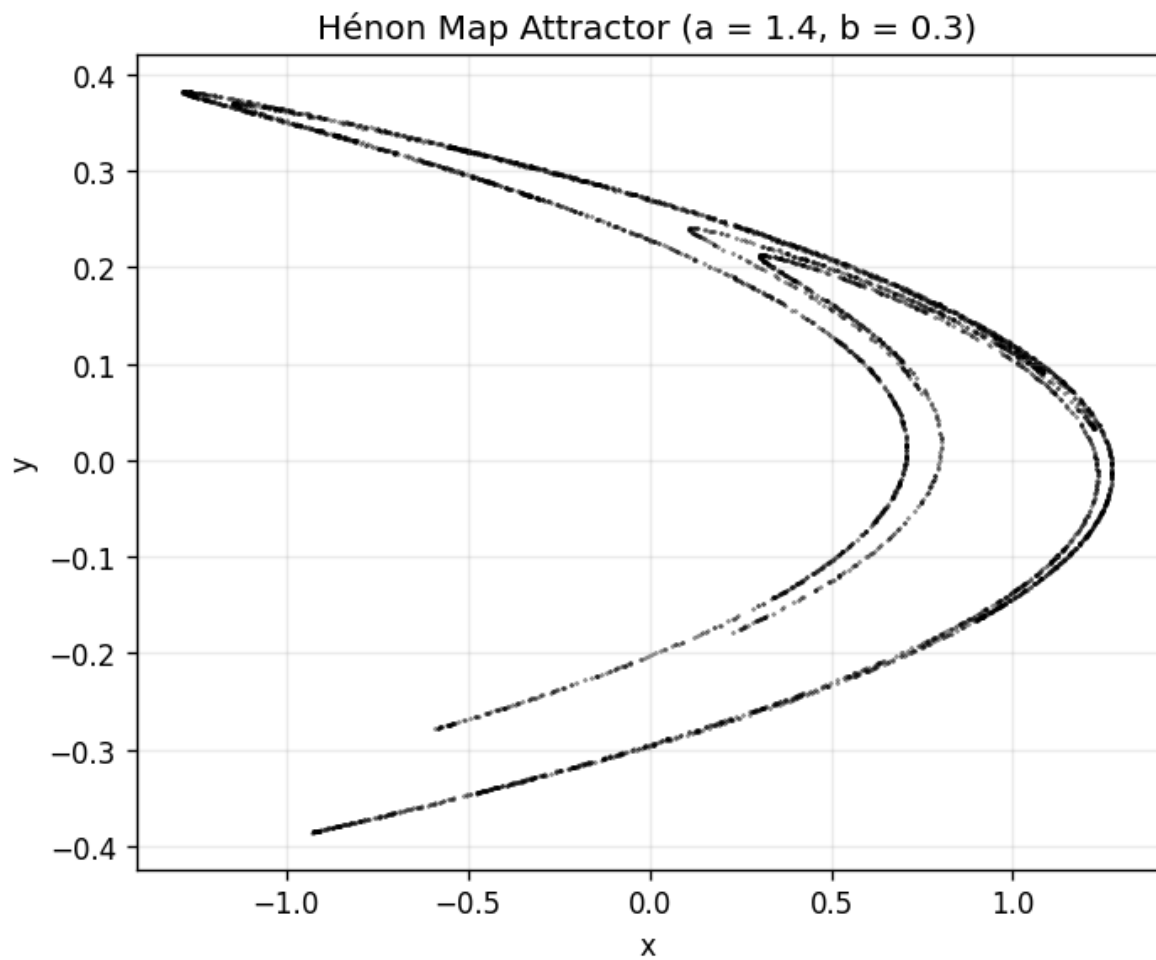
The bifurcation diagram shows how the long-term value of x changes when r changes. For smaller r values, the system settles to one stable value. Then it splits into two values, then four values, which is called “period-doubling cascade”, and later enters a chaotic region where many possible long-term values appear for the same r .

A.2 Hénon Map

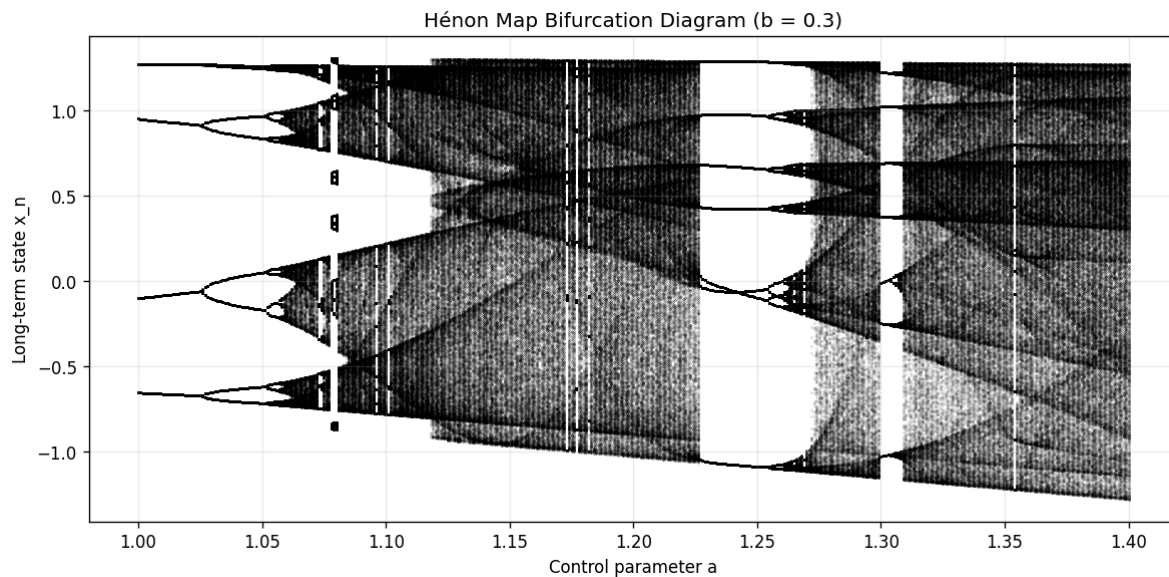
$$x[n+1] = 1 - a \cdot x[n]^2 + y[n]$$

$$y[n+1] = b \cdot x[n]$$

Setting	Value
State variables	x, y
Control parameters	a, b
Classic parameters	$a = 1.4, b = 0.3$
Initial condition	$(x_0, y_0) = (0, 0)$
Attractor iterations	5000
Transient ignored	first 500 iterations
Bifurcation range	a from 1.0 to 1.4, with $b = 0.3$ fixed
Bifurcation step	0.001
Bifurcation iterations	3000 total, first 1000 ignored



The attractor is the set of points that the system keeps visiting after the transient part disappears. The points stay inside a bounded region and form a folded structure instead of filling the whole plane.



From the diagram, the diagram has 4 dots and then almost doubles and then spreads into wider bands and enter chaotic behaviour at almost ($a = 1.6$). and then for some 'a' values, the system settles again into a small number of repeated values at ($a = 1.25$ and $a = 1.3$)

Part B: Continuous Chaotic Systems

B.1 Lorenz System

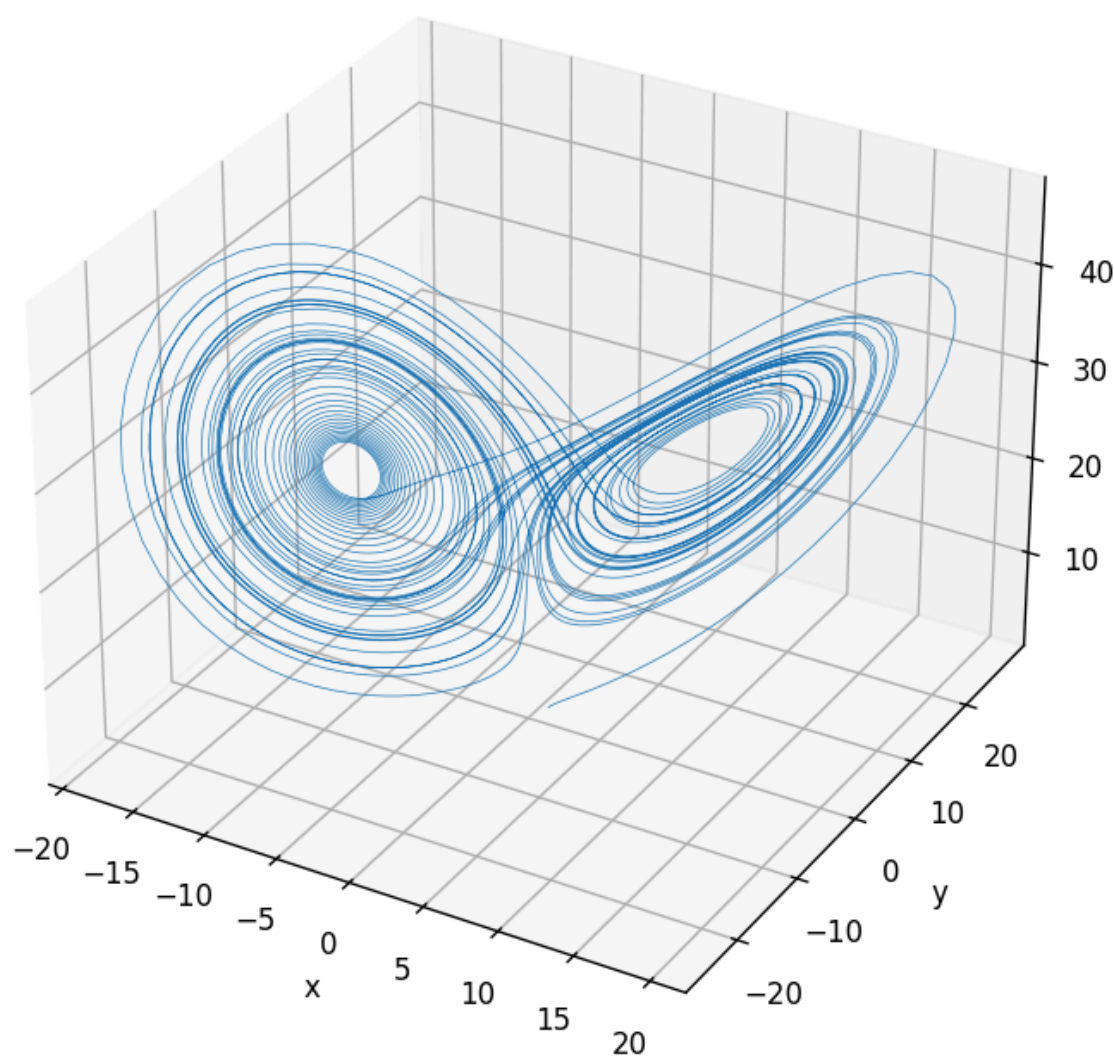
$$dx/dt = \text{sigma}*(y - x)$$

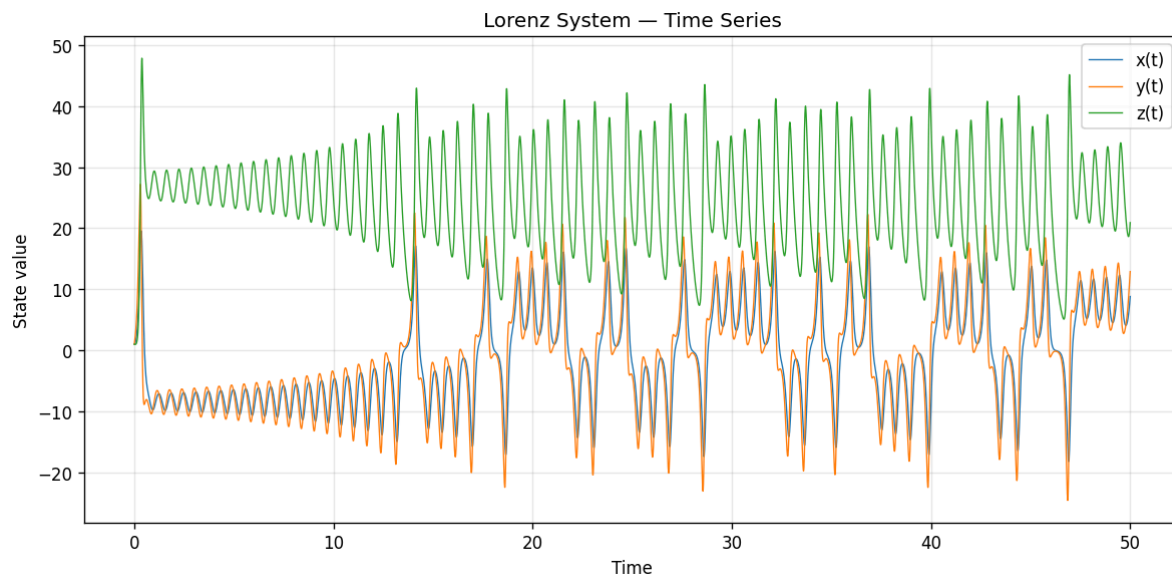
$$dy/dt = x*(\text{rho} - z) - y$$

$$dz/dt = x*y - \text{beta}*z$$

Setting	Value
State variables	x, y, z
Parameters	sigma = 10, rho = 28, beta = 8/3
Initial condition	(1, 1, 1)
Time step	dt = 0.01
Simulation length	T = 50
Solver	Fixed-step RK4

Lorenz System — 3D Phase-Space Trajectory





The Lorenz phase plot shows butterfly-like shape with two lobes. The trajectory keeps moving between both sides in an irregular way. The time-series plot also shows that x , y , and z are bounded but not periodic in a simple repeated pattern. This is a clear sign of chaotic behaviour.

B.2 Rössler System

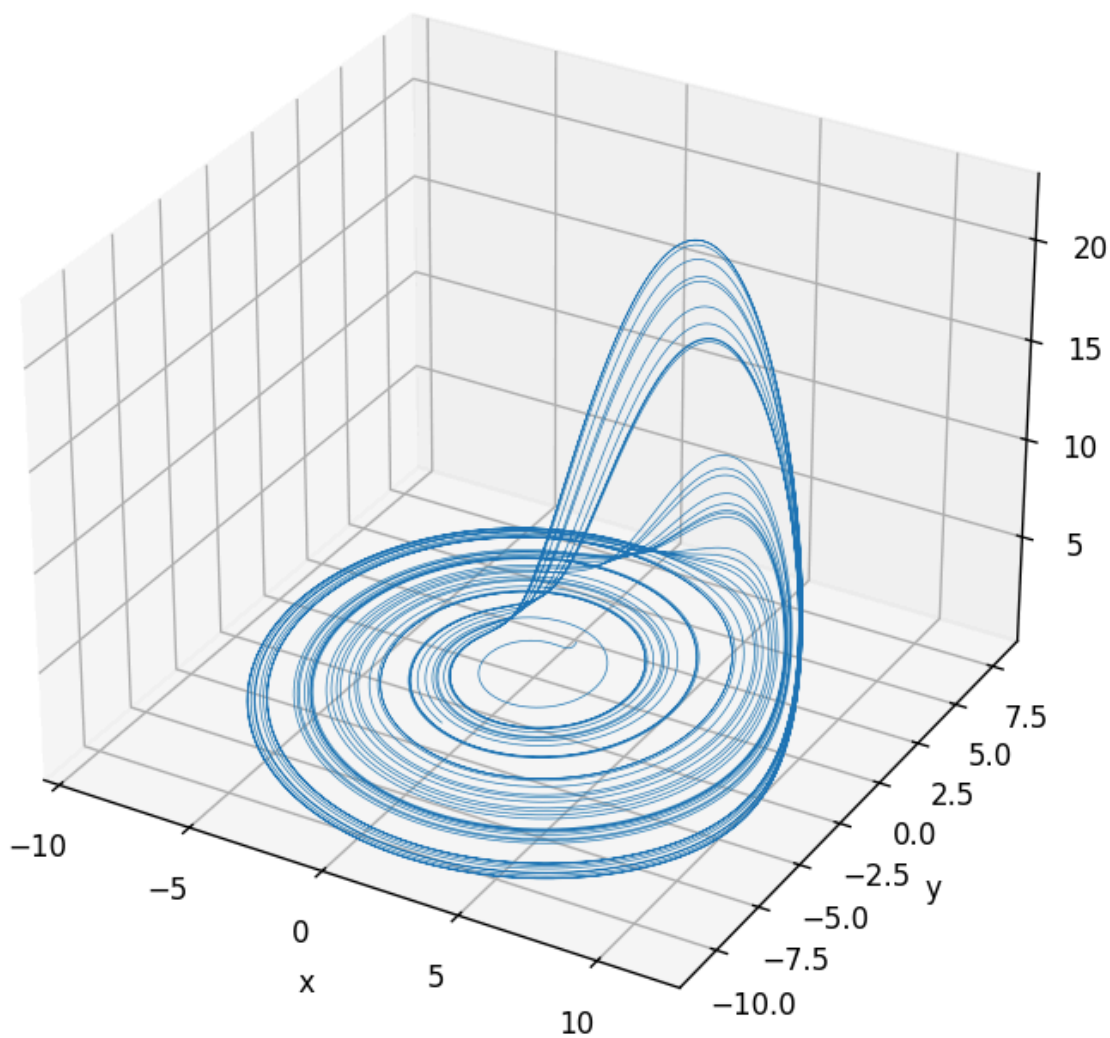
$$dx/dt = -y - z$$

$$dy/dt = x + a*y$$

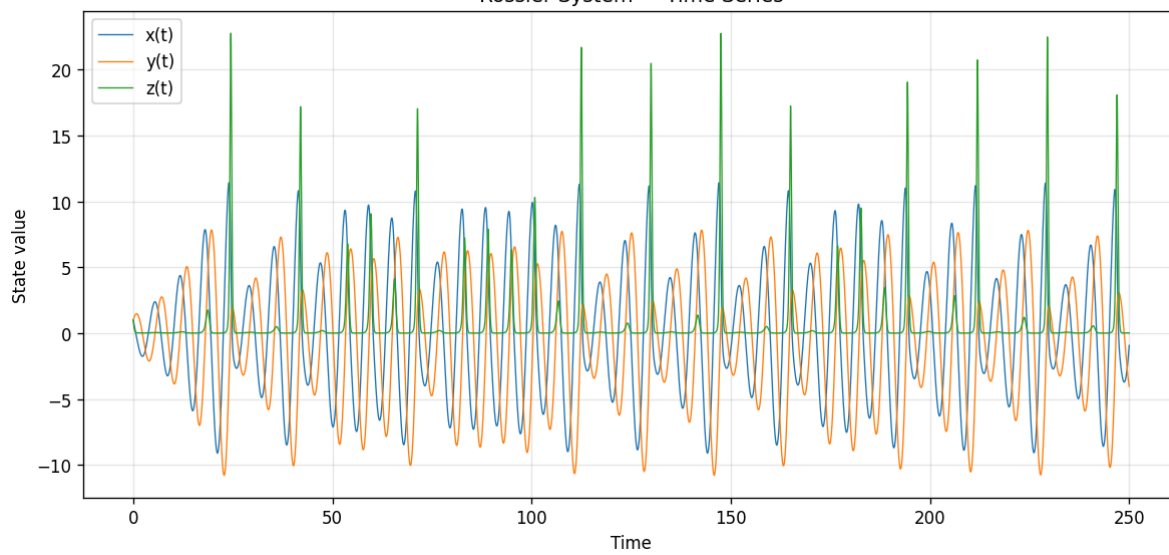
$$dz/dt = b + z*(x - c)$$

Setting	Value
State variables	x, y, z
Parameters	$a = 0.2, b = 0.2, c = 5.7$
Initial condition	$(1, 1, 1)$
Time step	$dt = 0.01$
Simulation length	$T = 250$
Solver	Fixed-step RK4

Rössler System — 3D Phase-Space Trajectory



Rössler System — Time Series



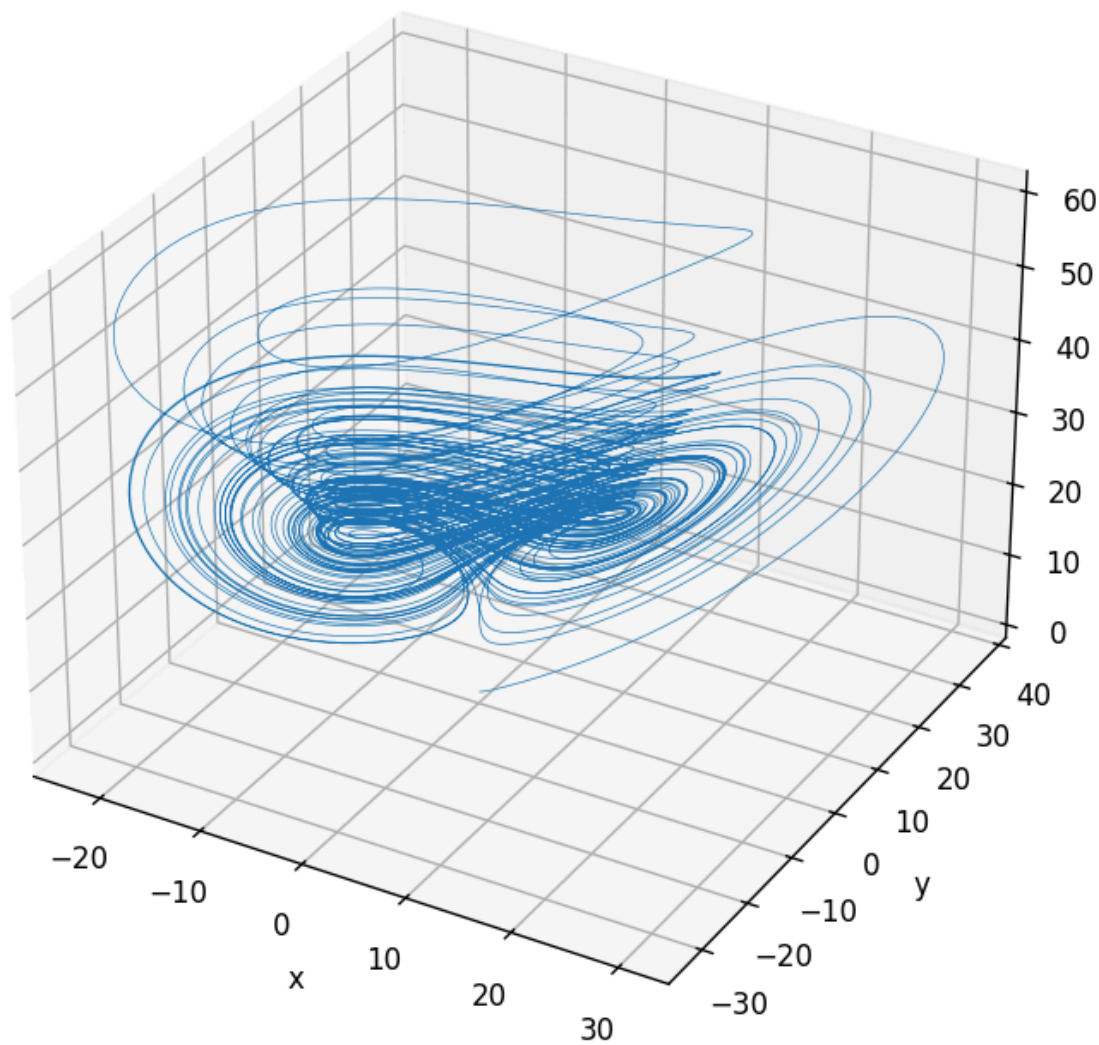
The Rössler attractor has a single-scroll spiral shape. The system rotates around the main spiral and then moves outward and back again. The time series shows repeated oscillations, but the peaks are not exactly identical, which indicates chaotic behavior.

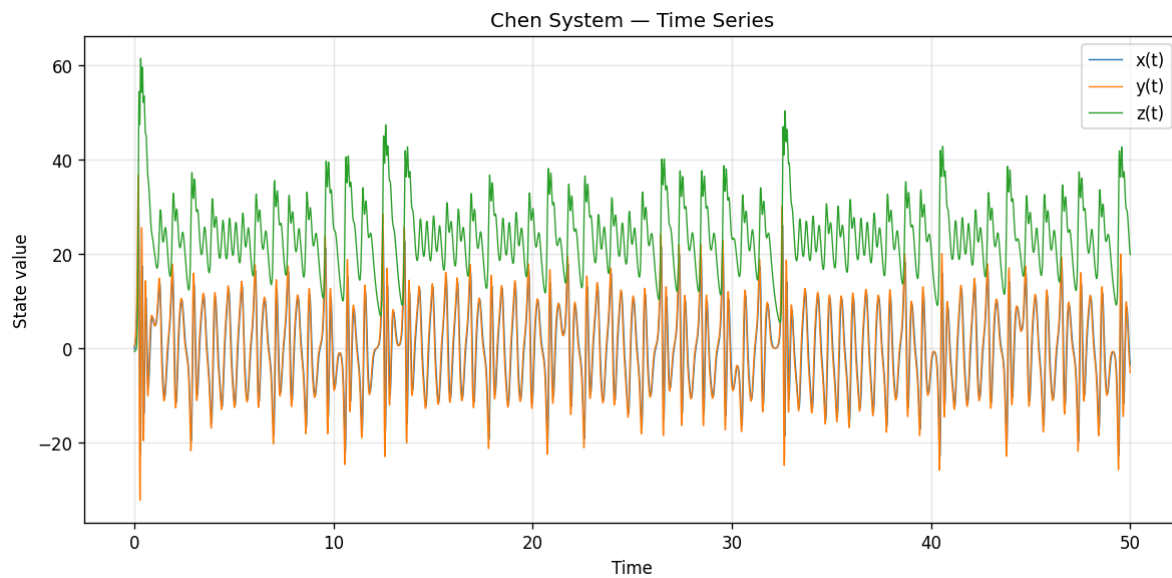
B.3 Chen System

$$\frac{dx}{dt} = a \cdot (y - x)$$
$$\frac{dy}{dt} = (c - a) \cdot x - x \cdot z + c \cdot y$$
$$\frac{dz}{dt} = x \cdot y - b \cdot z$$

Setting	Value
State variables	x, y, z
Parameters	a = 35, b = 3, c = 28
Initial condition	(-0.1, 0.5, -0.6)
Time step	dt = 0.002
Simulation length	T = 50
Solver	Fixed-step RK4

Chen System — 3D Phase-Space Trajectory





The Chen system creates a double-scroll attractor. It looks a little similar to the Lorenz system because the trajectory moves between two main regions, but the shape is different. The variables change irregularly with time while remaining bounded.

B.4 Hyperchaotic Rössler System

$$dx/dt = -y - z$$

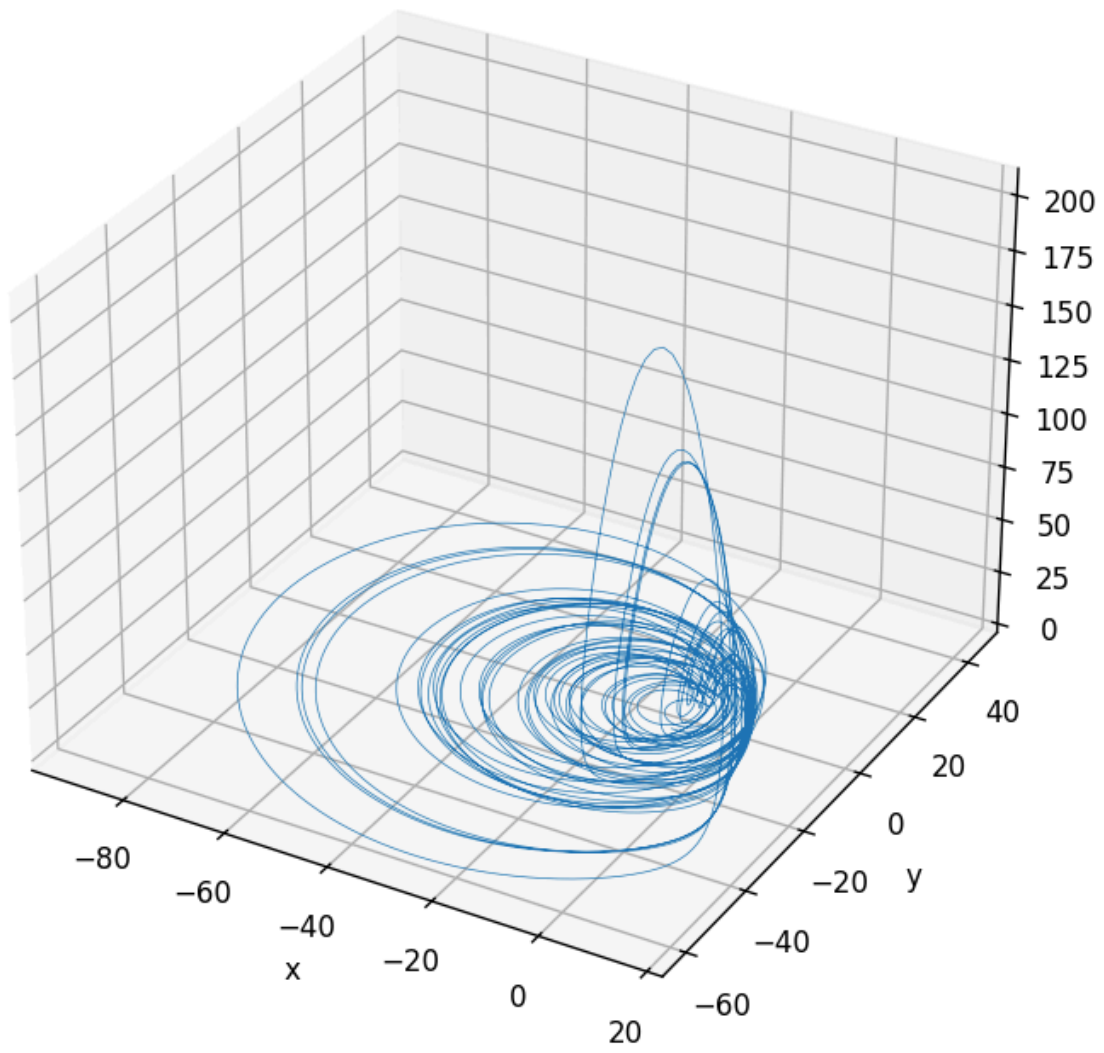
$$dy/dt = x + a*y + w$$

$$dz/dt = b + x*z$$

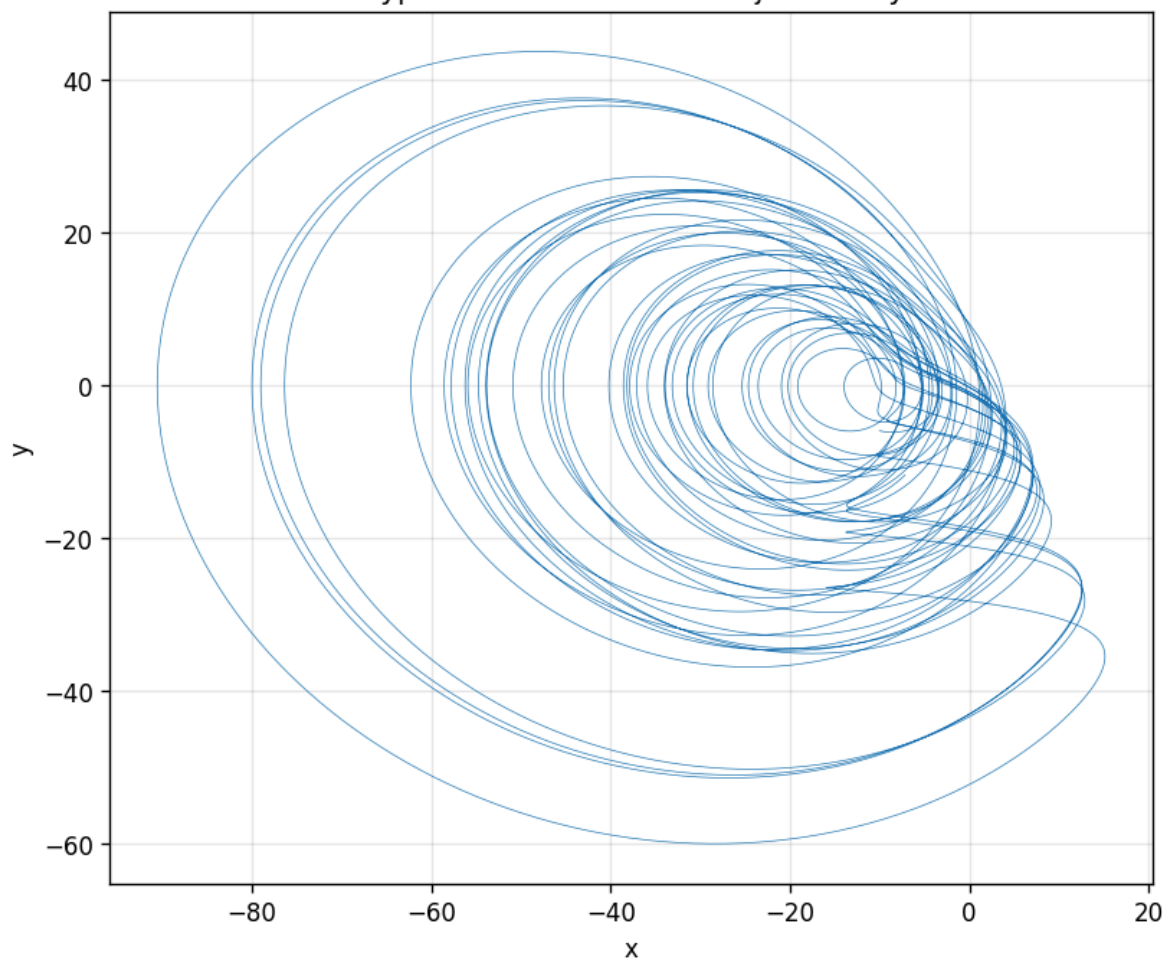
$$dw/dt = -c*z + d*w$$

Setting	Value
State variables	x, y, z, w
Parameters	$a = 0.25, b = 3, c = 0.5, d = 0.05$
Initial condition	$(-10, -6, 0, 10)$
Time step	$dt = 0.01$
Simulation length	$T = 250$
Solver	Fixed-step RK4
Plots used	3D and 2D projections because the full system is 4D

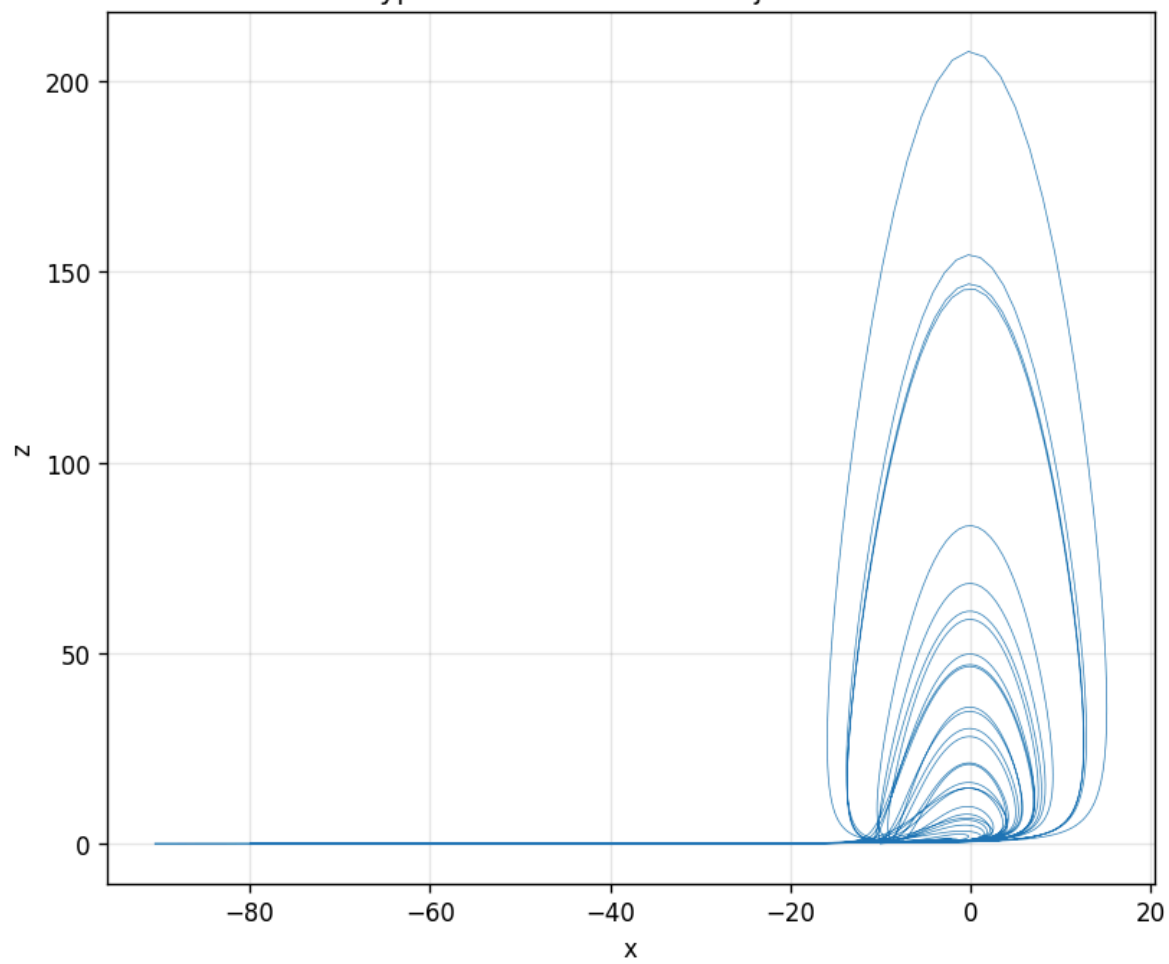
Hyperchaotic Rössler — 3D Projection (x, y, z)

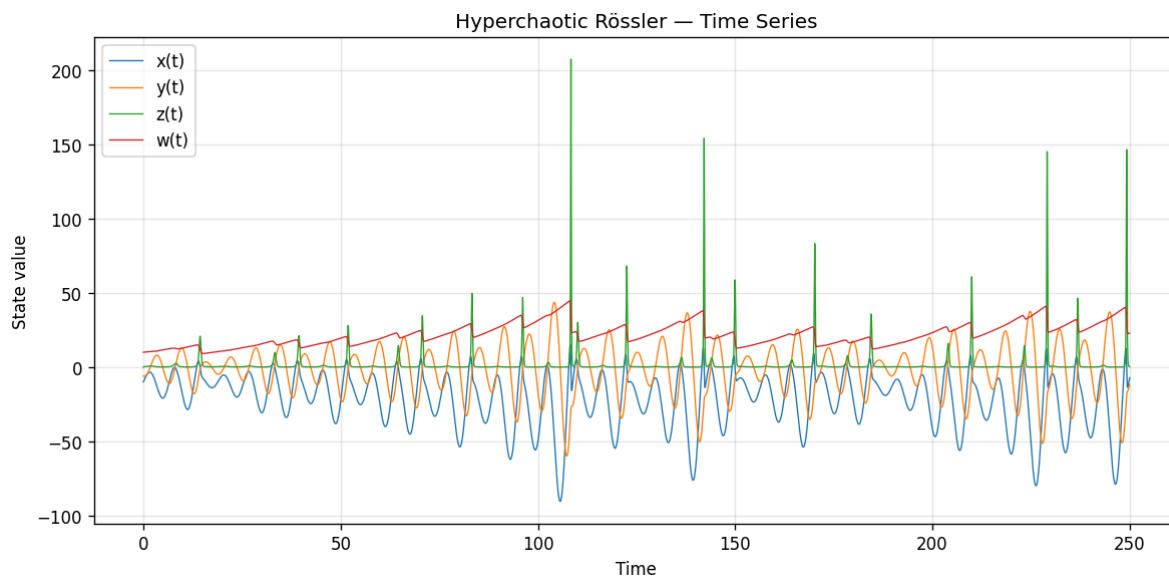
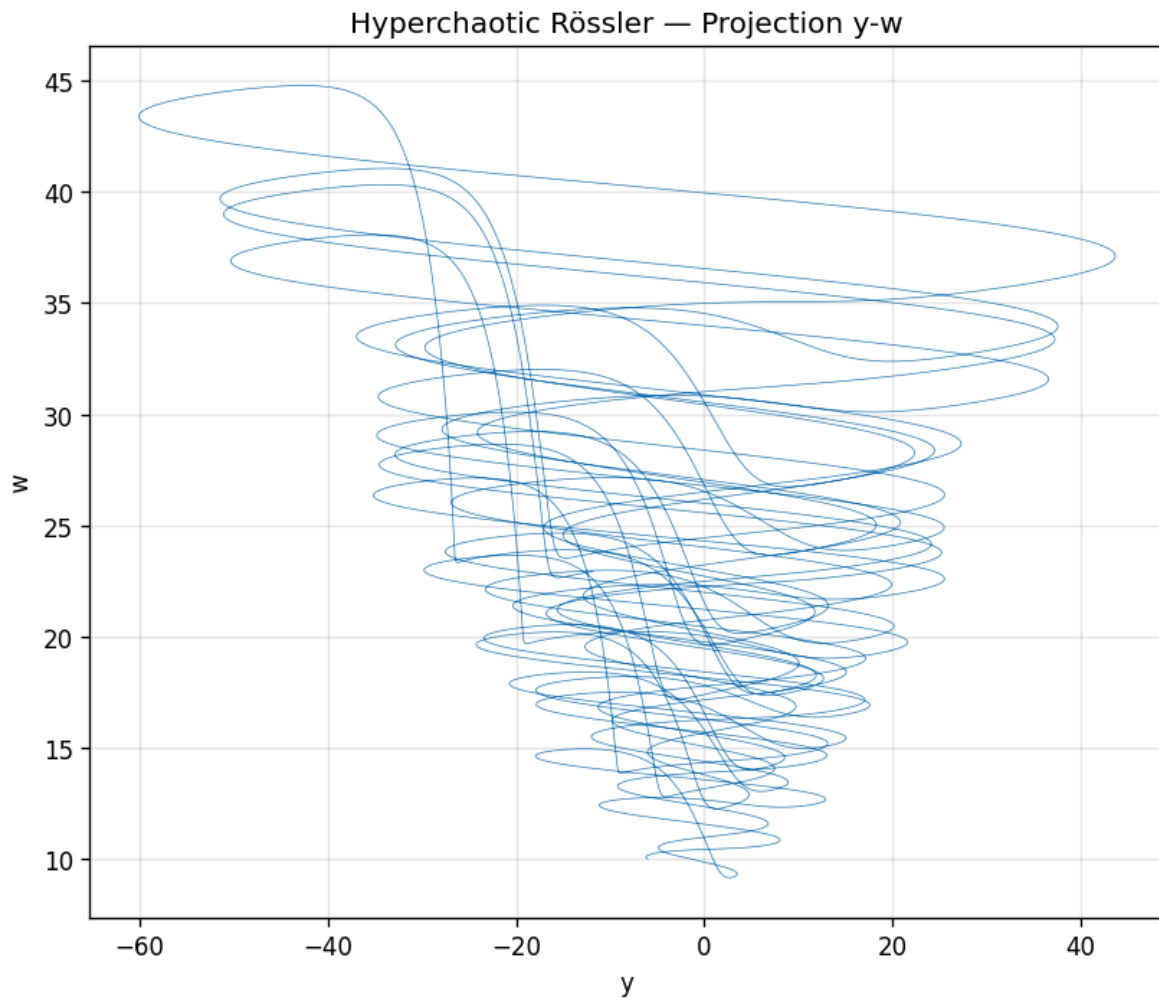


Hyperchaotic Rössler — Projection x-y



Hyperchaotic Rössler — Projection x-z





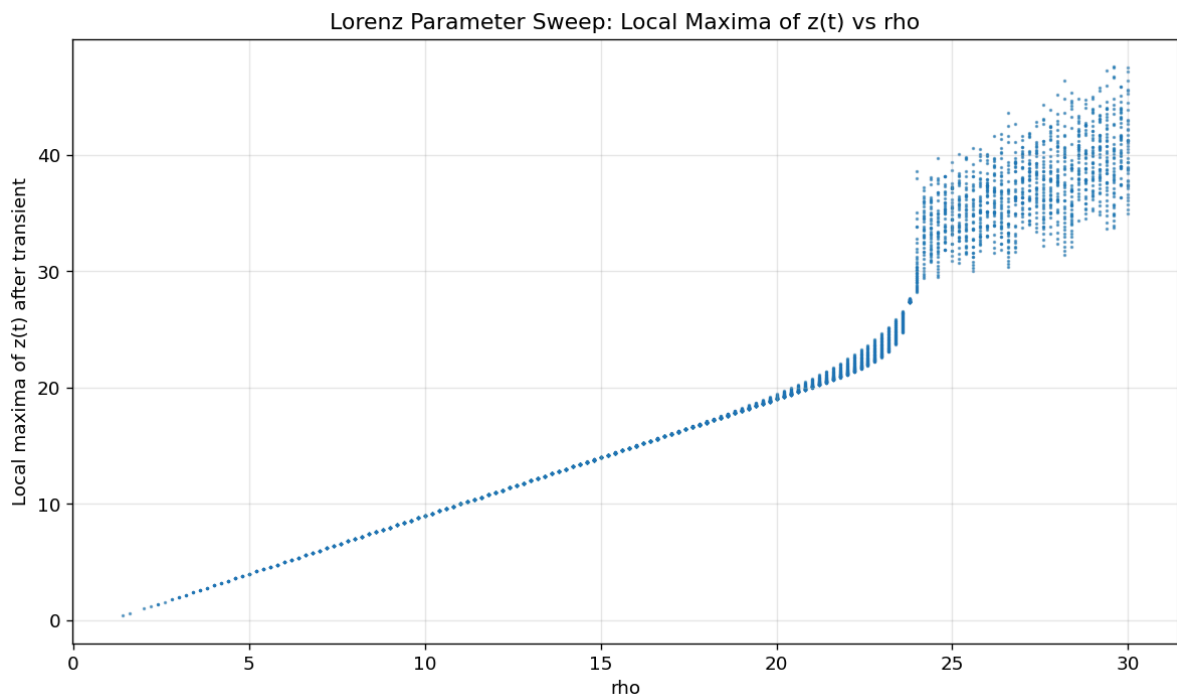
The hyperchaotic Rössler system has four state variables, so the full phase space cannot be drawn directly. Because of that, I used 2D and 3D projections. The projections show a more complex structure than the normal 3D systems. The time series also shows irregular bounded motion in all four variables.

Part C: Sensitivity Analysis

In this part, I studied two types of sensitivity in the Lorenz system. The first one is sensitivity to the control the parameter “rho”. The second one is sensitivity to the numerical time step dt.

C.1 Lorenz Parameter Sweep

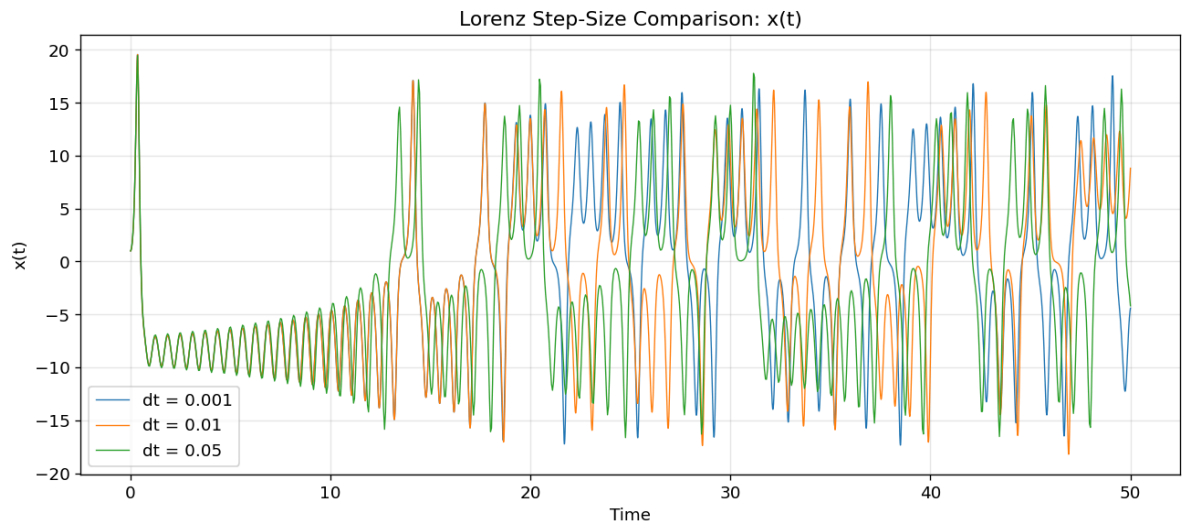
Setting	Value
Swept parameter	rho from 0 to 30
Step in rho	0.2
Fixed parameters	sigma = 10, beta = 8/3
Initial condition	(1, 1, 1)
Time step	dt = 0.01
Simulation length	T = 50
Transient ignored	first 20 time units
Recorded quantity	local maxima of z(t)
Solver	Fixed-step RK4



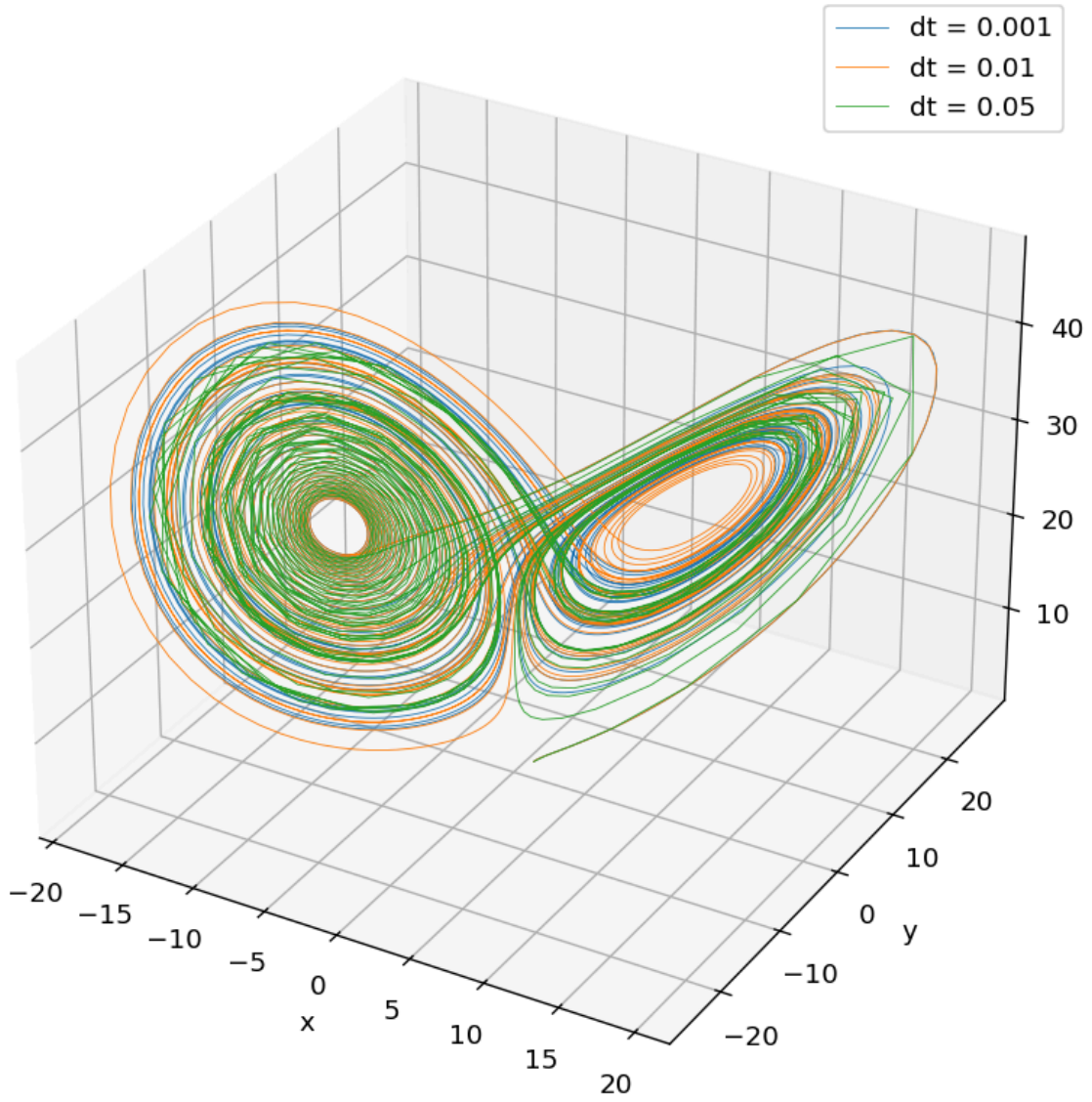
The plot shows that changing “rho” changes the long-term behavior of the Lorenz system. For lower “rho” values ($\rho < 20$), the maxima follow a simpler structure. As “rho” increases, the points become more spread and scattered. This indicates that the system becomes more complex and chaotic as the control parameter “rho” changes.

C.2 Step-Size Sensitivity

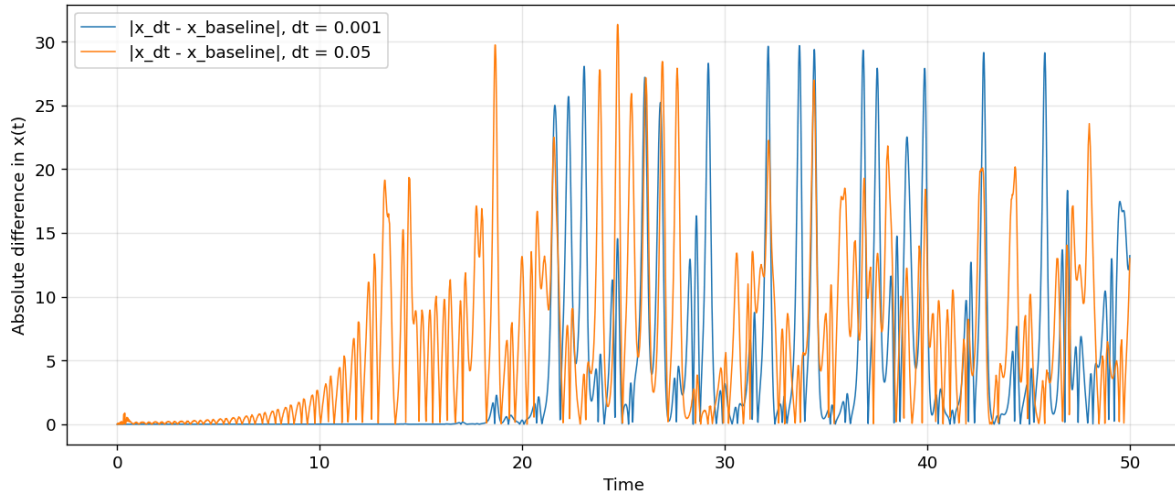
Setting	Value
System	Lorenz system
Parameters	$\sigma = 10$, $\rho = 28$, $\beta = 8/3$
Initial condition	(1, 1, 1)
Step sizes tested	$dt = 0.001, 0.01, 0.05$
Baseline step size	$dt = 0.01$
Simulation length	$T = 50$
Solver	Fixed-step RK4
Comparison used	$ x_{dt}(t) - x_{baseline}(t) $



Lorenz Step-Size Comparison: 3D Phase Space



Lorenz Step-Size Sensitivity: Divergence from Baseline



At the beginning, the trajectories are close to each other. After some time, they start to separate. The coarse time step $dt = 0.05$ is less accurate and can miss some details, while $dt = 0.001$ and $dt = 0.01$ give a better representation of the attractor.

Conclusion

From these simulations, I noticed that discrete and continuous chaotic systems can both produce irregular behavior from simple equations. The discrete systems move by iterations, while the continuous systems move with time using differential equations. In both cases, the behavior stays bounded but unpredictable.

The chaotic systems, like Lorenz, Rössler, and Chen, show sensitivity and strange attractors in 3D phase space. The hyperchaotic Rössler system is more complex because it has four state variables and can separate nearby trajectories in more than one direction. Overall, the results show that chaos is not just random motion. It comes from deterministic equations, but small changes in parameters, initial conditions, or numerical settings can lead to very different results.

Appendix: Simulation Code

The full simulation code is attached in the uploaded notebook. The notebook includes the Python functions used to generate the figures, map simulations, the RK4 solver and plotting functions.